

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

This month we delve into artificial nervous systems, tackle the age-old question "How old is the moon?", review a new form of "mind control" and explore Galactic weather.



How long does it take to make a friend?

A researcher investigating the amount of time it takes to forge friendship has specific numbers for us. It takes 50 hours of time spent together to make you a "casual friend", 90 hours to become a "friend", and over 200 hours to become "close friends".

<https://digit.in/frndhr>

WHAT'S NEW

Research shows that the so called "Nemo effect" is a myth

After the respective releases of "Finding Nemo" and "Finding Dory", reports said that the demand for clown fish and the blue tang fish had spiked around the world. One of the most high profile appeals to not buy the species was from the voice of Dory, Ellen DeGeneres. Researchers from the University of Oxford decided to check if there really was a change in the behaviour of pet buyers that could be linked to movies. By correlating sales of pets with Google trends, the researchers found out that there is no clear link. However, the movies do drive search interest. This results in an increase in awareness



about wildlife. Diogo Verissimo, the lead researcher of the study says, "I was intrigued as to whether the

connection between these blockbusters and demand for wildlife was as straight-forward as had been described in the media. My experience is that human behaviour is hard to influence at scale and it seemed unlikely that movies generated spikes in demand for the species they feature".

<https://digit.in/nemoeffect>

Robots learn jugaad

Researchers at Georgia Tech taught a robot to combine an assortment of objects into new tools. For example, the robot combined a coin and pliers to make screwdrivers. The robot analyses the shapes of the objects provided, and thinks about how they can best be combined together to perform a particular task. However, the robot still does not understand some concepts, such as weight, so it can try



to make a hammer out of the block of foam instead of a block of metal. The researchers are now attempting to teach the robot the properties of materials as well, and incorporate that into the logic of building new

tools. The inspiration for the initiative comes from a famous example where MacGyvering actually helped save the lives on board the Apollo 13 mission. An oxygen tank blew out on the way to the Moon and the mission had to be immediately aborted. The crew had to fix the carbon dioxide removal system using a makeshift filter. The team at Georgia Tech hopes to prepare robots for such contingencies.

<https://digit.in/robotjugaad>



Viviparous lizard extinction

Research shows that lizard species that bear young evolved the ability to adapt to cold climates. Because of this, the lizard species were trapped in colder regions, in higher altitudes and latitudes. With global warming, these lizards may go extinct within the next 60 years.

<https://digit.in/coliz>



The canine cancer parasite

Researchers are studying a canine cancer parasite that spread from a single host 1900 years ago, to dogs around the world, including those living on islands. This indicates that the parasite hitched a ride on ships. The cells of the tumour belong to the original dog, and not the host dog.

<https://digit.in/cancan>



Pac-Man black hole

Researchers discovered a black hole is going to swallow a neutron star. While the collision of two black holes, and two neutron stars have been detected, it is the first time that a black hole swallowing a neutron star has been detected, completing the trifecta for gravitational wave observing instruments.

<https://digit.in/nsbh>